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Circular economy and nature-based solutions for WASH resilience in Bangladesh's Hill tracts

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Abstract

Climate change is one of the pressing issues in Bangladesh, a developing country in South Asia. Reliable access to safe water and adequate sanitation remains a critical challenge for indigenous communities in this country's southern part, i.e., Chittagong Hill Tracts (CHT), where steep terrain and seasonal climate shocks intensify fetching burdens and health risks. They are mostly dependent on natural resources to meet their basic needs. A few studies are addressing their Water, Sanitation, and Hygiene (WASH) condition and climate vulnerability, as well as nature-based solutions with a focus on the circular economy. With that view, a cross-sectional survey was conducted on 81 households across four Chakma, Tripura, and Bangali settlements. Evaluating the perception and effect of climate change on the underprivileged people of CHT revealed a bitter truth. Farmers are leaving paddy production to minimize water use in agriculture and investing money with high interest to afford the initial cost of Tobacco cultivation. Consequently, food scarcity and financial burden make their lives more difficult, with prevailing water, sanitation, and flood problems. Descriptive analyses revealed average round-trip fetching times of 7.8 min for drinking water and 19.4 min for domestic water, and 60% of households experience seasonal or chronic scarcity. One-way ANOVA and multiple regression showed that each additional 100 m distance to a water source increases domestic-fetching time by 3.1 min ($p < 0.001$), while higher climate-shock indices add 1.8 min ($p = 0.003$). Principal Component Analysis (PCA) and cluster analyses identified a high-burden profile encompassing one-third of households. These findings urge targeted WASH investments and align with SDG 6 and climate-resilience goals. Therefore, this study concluded by incorporating a circular economy to strengthen governance and financial systems through nature-based solutions. Integrating proper health, sanitation, and technological knowledge with interest-free investment in the communities can be a sustainable solution.

Keywords Circular economy, Governance and financial system, Nature-based solutions, WASH, Underserved communities

1 Introduction

Rural CHT communities of Bangladesh face chronic shortages of WASH facilities, which are exacerbated by challenging topography, deforestation, and rising climate variability. These synergistic stressors increase daily water-collection challenges, particularly



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among women and children, and worsen water quality during floods, and limit livelihood opportunities, creating a highly interdependent socio-ecological weakness that can be hard to overcome with only infrastructure investments [1–4]. Nature-based interventions (NBS) like riparian reforestation and hill-slope wetland restoration can stabilize springs and recharge aquifers, and documents like rainwater harvesting, organic waste composting, and greywater treatment are examples of measures at the circular-economy scale that provide opportunities to reduce household-level loads and create local value; but there is little evidence as to how household-level burdens relate to priority [5–8].

Current literature portrays existing WASH deficits and climatic effects in the circumstances of Bangladesh coastal, urban, and hill, yet these are piecemeal: various reports describe the issue, or pilot single technologies, and scarcer ones combine rigid household-level quantitative measurements (e.g., fetching time, adequacy odds) alongside system-level governance and financing arrangements that would allow scale and equity [9–13]. Past studies tasked with CHT, in particular, emphasise the relationship between forest and water shortage, seasonal dependency on springs and streams, and the gendered collection burden, but offer few comparative studies, which integrate these domestic demands with specific circular-economy policies and governance strategies [1, 2, 14, 15]. This is a gap that restricts the capacity of policymakers to make investments both to mitigate the daily burden and enhance resilience in the long run.

In this research, the overall aim is to examine how water access, sanitation, livelihoods, and socio-economic well-being in four remote villages (Chakbachara, Betchari, Pakizachari, Dighinala), Dighinala Upazila, Khagrachhari, have changed due to climate-induced variations in rainfall, water availability, and extreme events, and what locally feasible circular-economy interventions are. Besides, this research also sought to:

- (a) Examine existing water-collection burdens: distance, round-trip time, quantity to drink, and household use, and compare to pre-2000 surface water dependency.
- (b) Sanitation coverage, open-defecation levels, and health-hygiene knowledge deficits survey and attribute them to water-access limitations and water-driven pollution by floods.
- (c) Examine the prevalence and perceived severity of climate changes (droughts, flash floods, landslides), and assess their effects on conventional paddy as well as jhum cultivation, income volatility and indebtedness, and migration intentions.
- (d) The community awareness of climate change, adaptive patterns (crop shifts, use of infrastructure), and overcoming barriers to adopting technology (solar pumps, groundwater extraction).
- (e) Energize stakeholder-led suggestions, such as compost-latrines-based circular nutrient utilization, groundwater pumping, basic water-filtration lessons, and zero-interest credit plans to educate a participatory WASH-resilience map to poor hill-tract neighbourhoods.

The methodological approach of the study is based on a stratified cross-sectional survey of households and primary water-fetchers in four hill villages to measure round-trip fetching times, distance, sanitation coverage, and climate-shock exposure, and subsequently relate the empirical findings to intervention design criteria (cost, reliability, co-benefits, and governance fit). In addition, Focal Group Discussion (FGD) and Key Informants Interview (KII) were conducted to reveal socio-economic conditions. The

analysis should provide actionable evidence, which can lead to specific investments, i.e., not general advice, to make interventions as efficient as possible to target households needing them the most and having the highest probability of reaping co-benefits.

The piece of work has a two-fold contribution. It offers an empirically-based, household-based, statistically-grounded evidence on the cause of WASH burden in an extremely marginalized hill-tract environment; information explains the comparative significance of distance and climate shocks, sanitation ownership, and livelihood dependency. In practice, it applies these findings to a governance-conscious roadmap of prioritized, circular-economy- and nature-based WASH interventions, including community-level rainwater harvesting, compost-latrines and interest-free financing models, which national adaptation objectives and SDG 6 should incorporate, though in a politically and institutionally contextualised manner.

2 Methodology

The step-by-step methodology has been explained below, and at a glance, it is demonstrated in Fig. 1.

2.1 Study design and sampling

The cross-sectional survey was conducted from March to May 2025 across four hill-tract villages in southeastern Bangladesh (Chakbachora, Betchari, Pakajjachori, Dighinala). A total of 160 individual interviews had been completed with 81 households; the household records were used to analyze the household-level data, with the interview records being used to analyze the data at the individual level (e.g., ANOVA on respondent fetching times). The criteria the respondents had to meet were being 18 years of age, being a permanent resident (not less than 1 year), and being the main water-gatherer in their household. The stratified random sampling of households was based on village population, and within sampled households, one or two primary water-fetchers were chosen, with the interview being conducted with them where they were available, giving rise to the 160 individual interviews. Despite the small sample employed (81 households, which represent over a third of the whole population in four villages; on average, 60 households per village emerged in the survey based on FGD), a proportional coverage of the entire population was achieved, and the ethnic identity, livelihood dependency, and remoteness were well covered. The stratified random sample design increases representativeness,

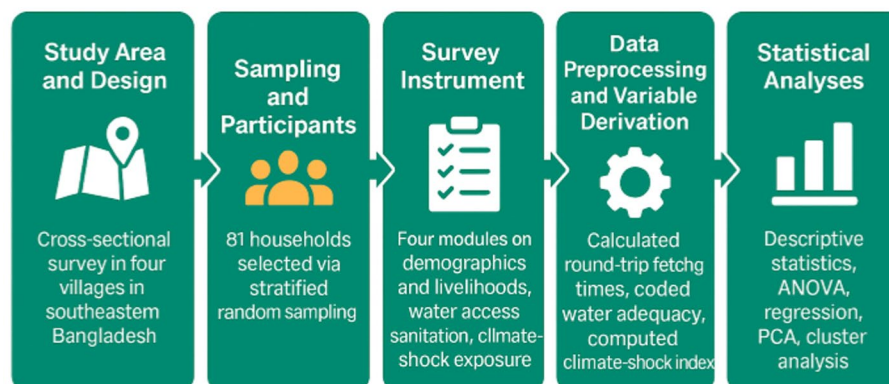


Fig. 1 Methodology at a glance

and hence the results can be generalized to other vulnerable hill communities in Bangladesh, where a small population, lack of government support, and being in dire climatic conditions are comparable resilience changes. FGD was conducted with a group of 5 aged people (permanent residents) from each village, capable of describing the socio-economic conditions of the previous century. KII has captured educational, agricultural, and financial pictures of the areas from an overview of school teachers, heads of the society, and some prominent farmers. Figure 2 documents some of the data collection scenarios.

2.2 Survey instrument and modules

The semi-structured questionnaire maintained the four modules: (a) demographics and livelihoods; (b) water-access measures (types of sources, round-trip fetching time, round-trip distances, perceived adequacy); (c) sanitation facilities, and (d) exposure and adaptation to climate shocks. Start-to-return timestamps that were rounded off were used as round-trip fetching times. The adequacy of water was rated on a three-level ordinal scale (1 = always sufficient; 2 = seasonal crisis; 3 = chronic crisis). The climate shock index is the aggregate of self-reported droughts, cyclones, and landslides in the last five years. The fraction of household income on agriculture is referred to as agriculture-dependency. FGD captured the WASH condition and the socio-economic conditions of the previous century and compared them with the next phases. Population, education, historic event along with socio-economic crisis were found from KII.

2.3 Data preprocessing and analytic samples

Drinking and domestic round-trip time (minutes), distance to drinking and domestic sources (meters), and standardized continuous predictors (z-scores) were derived variables. The household dataset was analyzed to describe household infrastructure, PCA, and cluster analysis. The individual interview dataset was analyzed to study individual respondent results (e.g., ethnic-group differences in reported domestic fetching time). In all the tables and captions of figures, it is clearly mentioned which sample ($n=81$ households or $n=160$ individuals) is used. Drought exposures (past 5 years) + cyclone exposures (past 5 years) + landslide exposures (past 5 years) = climate-stock index. Agriculture-dependency ratio (percent income from agriculture/100). Predictors that were continuous were normalized (mean = 0, SD = 1) to interpret the coefficients easily.

2.4 Statistical analyses

All analyses were performed in R (v4.2.2) with packages 'stats', 'MASS', 'FactoMineR', and 'cluster'.

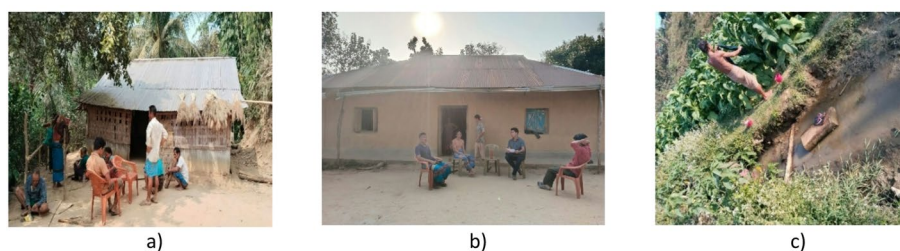


Fig. 2 Field survey **a** FGD **b** KII **c** domestic water collector

- (a) Descriptive Statistics: The most important variables were summarized by means, standard deviations, and proportions.
- (b) One-Way ANOVA: Evaluated the difference between ethnic groups in domestic-fetching time; Tukey's-HSD in pairwise contrasts.
- (c) Multiple Linear Regression: domestic-fetching time; predictors: distance, agriculture dependency, climate-shock index, ownership of toilets, ethnic dummies.
- (d) Ordinal Logistic Regression: ordinal scale of water-adequacy; same predictors as linear regression; proportional-odds test enforced with Brant test.
- (e) PCA: Indicator Set: drinking/domestic fetching times, distances, adequacy score, toilet ownership; retained component with eigenvalues larger than 1.
- (f) Cluster Analysis: PC1/PC2 hierarchical clustering (Ward's method) to support household vulnerability profiles; silhouette width compared the stability of clusters.

The level of statistical significance was defined ($\alpha = 0.05$). An approximation of the variance-inflation factor (< 2.5) in multicollinearity and a plot of data (residuals) to test the conditions of normality and homoscedasticity constituted model diagnostics.

3 Results

3.1 Findings from FGD and KI

Mainly Chakma and a few Bengalis reside in villages, namely, Chakbachara (beside Myinee River), Betchari, Pakizachari (people who live on Tilla), and Dighinala (rural area). They generally can't afford electricity individually. Some of them use solar panels, while others combine electricity-operated pumps with surface water irrigation. After 1980, people from other areas migrated to these places. Therefore, deforestation and other human activities resulted in the early 2000s. People used to gain a 180% return on paddy cultivation at that time. In Jhum cultivation, the return is still 170%, but it is losing popularity due to its tediousness. Previously, their main crop production was paddy (Fig. 3d), which needed two months of continuous supply of water. Due to climate change, rainfall and water availability have decreased significantly (Fig. 3f), forcing these people to alter their crop cultivation practices. In addition, the establishment of tobacco factories (2004–2005) and less water consumption motivated them towards tobacco cultivation (Fig. 3e). From tobacco, they earn 2–3 lakhs per Bigha. They get money from Mahajan and employees of tobacco factories at very high interest; therefore, their

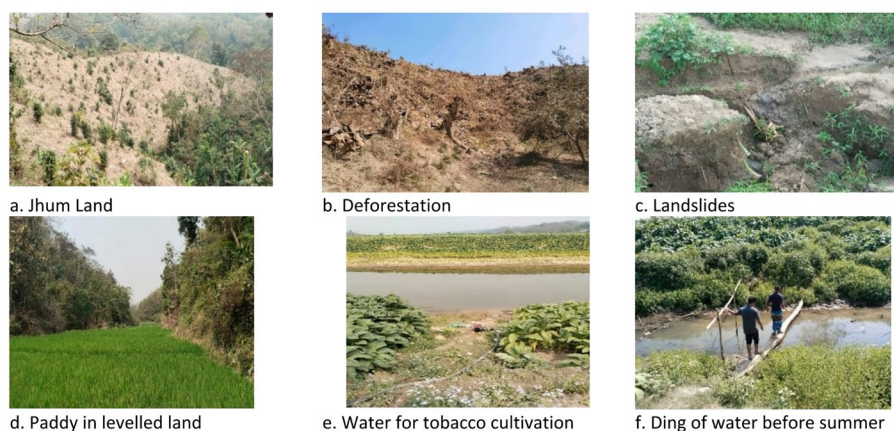


Fig. 3 Barriers to agriculture

**Fig. 4** WASH condition**Fig. 5** Field survey

economic condition never changed, even after gaining higher profit in tobacco than in paddy. Besides, the paddy production in Shifting cultivation (Jhum) is not effective as the same land can't be used next year (Fig. 3a), and cutting down trees for the preparation of land results in deforestation and landslides (Fig. 3b and c).

However, some of the lands are still being used for crops like paddy production to meet the demand for the staple food. The water from the mountains is carried to crop fields with narrow canals. But the main surface water dries up in summer.

Before 2000, the main source of water was a stream (chhora) and a canal (Khal) (Fig. 4a and d). The problem with surface water sources is the absence of water in summer and clayey water in the rainy season (the Canal gets submerged under water during flood). Groundwater is available within 50–60 feet (except Pakizachari, where there is black stone and no water, not even available in 300 feet). People use community pumps for collecting drinking water, but it is challenging due to the absence of reservoirs or storage facilities. The female, children, and elderly people face challenges in collecting water using steep roads, Fig. 4b. People bought a water tank with their own funds to store water in case of emergency (Fig. 4c).

Almost all sanitary facilities are at a significant distance from the water sources. In addition, they contaminate lower water sources during floods and rains. The area is deprived of a safe latrine for children and elderly people (Fig. 4e); only a single toilet was found in usable condition (Fig. 4f).

Natural disasters made existing infrastructure vulnerable, and even basic educational facilities are disrupted (Fig. 5a). However, people are trying to get a better life with the light of education, so they establish schools with local materials (Fig. 5b and c). The rate of attending secondary education is high, and few are highly educated. But they lack

Table 1 Summary of key socio-demographic and water-access indicators for the full sample (Households, $n=81$; Individual interviews, $n=160$)

Variable	Mean	SD	Range
Age (years)	43.8	15.2	18–95
Years in current location	6.4	8.7	1–120
Agriculture-dependency ratio	0.66	0.22	0.00–1.00
Drinking-water fetching time (min)	7.8	3.2	1–35
Domestic water fetching time (min)	19.4	7.5	5–60
Distance to drinking-water source (m)	85.6	40.3	< 25–250
Distance to domestic water source (m)	325.1	140.7	25–750
Private toilet ownership (%) (1 = yes; 0 = no)	60	–	0–1
Climate-shock index (events in 5 year) (count)	2.1	1.3	0–5

Table 2 Ethnic group-wise fetching time ($n=160$)

Ethnic group	<i>n</i>	Mean fetching time (min)	SD (min)
Chakma	104	20.3	7.8
Tripura	38	17.1	6.9
Bangali	18	14.8	5.5

basic knowledge of health and hygiene. They use muddy water for domestic purposes and don't bother to drink it. However, for health problems, they need to travel to distant medical facilities. Water-related diseases are common in the rainy season as their water sources become submerged under flood every year.

From the questionnaire survey among Chakma, Tripura, and Bangali, it was found that all respondents have been living there for more than 20 years and are mostly involved in the agricultural profession.

3.2 Sample characteristics and water-access metrics

Table 1 represents the ages of respondents' residence in the studied areas, ages, dependency on agriculture, distance of water sources and collection time, facilities of toilets, and lastly climatic event perceptions. Agricultural dependency is 0.66; the maximum water fetching time for drinking and domestic use is 35 and 60 min at distances of 250 and 750 m, respectively. Almost 60 households own toilets, but most of them are not in usable condition.

3.3 ANOVA: fetching time by ethnic group

One-way ANOVA tested the mean (Table 2) round-trip domestic fetching time across ethnic groups (Chakma, Tripura, Bangali). It revealed significant variation in domestic water fetching time by ethnic group ($F(2,157) = 5.62, p = 0.004$).

Post-hoc Tukey tests indicate Chakma households spend significantly more time than Bangali ($\Delta = 5.5$ min, $p = 0.003$); Tripura did not differ significantly from Bangali or Chakma at $\alpha = 0.05$. The overall mean domestic fetching time across respondents is 19.4 min (SD 7.5). One-way ANOVA on the 160 individual records indicates a small but statistically significant ethnic effect on reported domestic fetching time: $F(2,157) = 3.32, p = 0.038$. Tukey HSD pairwise contrasts show the Chakma–Bangali difference is the primary driver (mean difference ≈ 4.0 min, $p \approx 0.028$); other pairwise contrasts are not significant at $\alpha = 0.05$. ANOVA diagnostics (residual vs. fitted, Shapiro–Wilk on residuals, Levene's test) are summarized in Online Appendix A; residuals do not show severe

Table 3 Predictors of domestic fetching time ($n=81$)

Predictor	β (unstd.)	SE	t	p
(Intercept)	19.4	1.2	16.2	< 0.001
Distance (per 100 m)	3.1	0.4	7.8	< 0.001
Agriculture-dependency ratio	2.4	0.9	2.7	0.007
Climate-shock index	1.8	0.6	3.0	0.003
Toilet ownership (yes = 1)	− 2.9	1.1	− 2.6	0.01
Chakma (vs. Bangali)	3.8	1.3	2.9	0.004
Tripura (vs. Bangali)	1.9	1.5	1.3	0.194

$R^2 = 0.48$, adjusted $R^2 = 0.45$

Table 4 Ordinal logistic regression ($n=81$)

Predictor	OR	95% CI	z	p
Distance (per 100 m)	0.42	0.30–0.58	− 5.1	< 0.001
Agriculture-dependency ratio	0.55	0.36–0.85	− 2.6	0.009
Climate-shock index	0.63	0.45–0.88	− 2.8	0.005
Toilet ownership (yes = 1)	2.7	1.7–4.5	4.3	< 0.001
Chakma (vs. Bangali)	0.48	0.28–0.82	− 2.7	0.007
Tripura (vs. Bangali)	0.73	0.39–1.37	− 0.8	0.412

departures from normality, and homogeneity of variance is acceptable for the reported test.

3.4 Multiple linear regression: predictors of domestic fetching time

Domestic fetching time (minutes) (Table 3) was regressed on distance, agriculture-dependency, climate-shock index, toilet ownership, and ethnic dummies (reference = Bangali). The model explains 48% of the variance (adjusted $R^2 = 0.45$).

Each additional 100 m in distance increases the fetching time by about 3.1 min. A one-unit rise in the climate-shock index adds about 1.8 min. Chakma households spend significantly more time than Bangali households. Owning a private toilet reduces domestic fetching time by approximately 2.9 min. Model diagnostics: VIFs < 2.5 for all predictors; Shapiro–Wilk on residuals $p=0.12$ (no strong departure from normality); Breusch–Pagan test $p=0.21$ (no strong evidence of heteroscedasticity). Full regression table and diagnostic plots are provided in Online Appendix A.

3.5 Ordinal logistic regression: water-adequacy

An ordinal logit model predicted the odds of greater domestic water adequacy (1 = Always adequate → 3 = Chronic crisis) using the same predictors.

From Table 4, greater distance, higher agriculture-dependency, and more climate shocks decrease the odds of adequate domestic water. Toilet ownership more than doubles the odds of “Always adequate” status. Chakma households face significantly lower adequacy odds than Bangali households. Analysis of household and community predictors revealed several significant factors influencing WASH resilience in the CHT. Greater distance to water sources substantially reduced resilience, with households located 100 m farther away being less than half as likely to maintain reliable access ($OR=0.42$, 95% CI 0.30–0.58, $p<0.001$). Similarly, communities with higher agriculture-dependency ratios showed diminished resilience ($OR=0.55$, 95% CI 0.36–0.85, $p=0.009$), reflecting the vulnerability of livelihoods tied closely to climate variability. Exposure to recurrent climate shocks also lowered resilience odds by 37% ($OR=0.63$, 95% CI

Table 5 PCA ($n=81$)

Component	Eigenvalue	Variance (%)	Cumulative (%)
PC1 (Access)	2.45	49	49
PC2 (Sanitation)	1.12	22	71
PC3	0.78	16	87

Table 6 Cluster analysis ($n=160$)

Cluster	<i>n</i>	Mean PC1	Mean PC2	Description
Low-burden	48	− 0.82	− 0.31	Short fetching, private toilets
Moderate	92	0.10	0.04	Mixed sources, occasional crisis
High-burden	30	1.05	0.58	Long fetching, shared/open defecation

0.45–0.88, $p=0.005$). In contrast, toilet ownership emerged as a strong protective factor, with households possessing sanitation facilities being nearly three times more likely to demonstrate resilience ($OR=2.7$, 95% CI 1.7–4.5, $p<0.001$). Ethnic disparities were evident: Chakma households had about half the odds of resilience compared to Bangali households ($OR=0.48$, 95% CI 0.28–0.82, $p=0.007$), while Tripura households showed lower odds but without statistical significance ($OR=0.73$, 95% CI 0.39–1.37, $p=0.412$). These findings underscore the combined influence of geographic, socio-economic, and cultural factors in shaping WASH resilience outcomes. The proportional-odds assumption was tested (nominal test) and did not reject proportionality at conventional levels; full test statistics are in Online Appendix A.

3.6 Principal component analysis (PCA) of water-sanitation burden

PCA on five standardized indicators (drinking- and domestic-fetching times, distance, water-adequacy score, toilet-ownership [0/1]) yielded:

PC1 loadings fetching times (0.64–0.72), distance (0.68), adequacy (−0.70) – a composite “access-burden” axis. PC1 “Access burden,” and PC2 “Sanitation burden,” explaining 71% of the variance. Hierarchical clustering on PC1 and PC2 identified three vulnerability profiles from Table 5.

3.7 Cluster analysis: household vulnerability profiles

Hierarchical clustering on PC1 and PC2 scores (Ward’s method) identified three profiles:

Two-thirds of the population falls into the high-burden to Moderate profile (Table 6), signaling an urgent need for targeted WASH interventions.

3.8 Findings from the rest others questions of individual interviews

Climate change is not clearly understood by many, yet they can experience the temperature and the pattern of rainfall. Drought has been experienced by nearly everybody (more than once over the past few years); cyclones are also not uncommon here. Conversely, two-thirds of the population had lost their property as a result of landslides. 50% of the population is willing to change their occupation and residence due to climate problems. A few of them go to seek garment work or business in tourist destinations, and many are afraid of migrating. Government and NGO support is negligible. This research identified fewer than 10 individuals who did not experience any financial loss in the recent past because of climate problems. Likewise, 5% of the population reports

that their water issue is resolved by pumping and storage tank measures. Fewer than one-third believe that agriculture is economically profitable, yet over 60% of their overall earnings rely on agriculture. On average, a family of 4–6 people uses 10–30 L of drinking water and 40–80 L of domestic water. Mostly, women have to travel as far as 30 min and a distance of 500 m to collect water. In addition, One-third of the population defecates openly.

4 Discussion

The interview outcome has established that distance to water source is the most significant determinant of time burden and perceived adequacy, which is in line with WHO/JMP standards that indicate that > 30 min is considered as unimproved access. The exposure to climate shocks alone enhances the fetching times and reduces adequacy, which depicts the joint susceptibility of these hill ecosystems. Toilet ownership not only fills the sanitation gaps, but it also indirectly lowers the domestic water-collection times, probably by decreasing the need to use distant communal latrines or open defecation points, which necessitate a separate water trip. The presence of ethnic differences, especially between Chakma families, highlights the need to consider the cultural considerations in infrastructure planning and equitable distributions of resources. Cluster profiling presents a viable approach to prioritizing the 30% of households in the high-burden segment to be immediately supported. These interventions are to co-locate water points within 100 m ranges of dwelling clusters with small reservoir systems and solar-powered pumps to guarantee reliability throughout the year. The findings contribute to the current body of literature on rural WASH in hilly geographies by measuring the interactive effects of spatial distance, climate shocks, and sanitation status on measures of access. They also confirm the effectiveness of localized tube-well installations with regard to enhancing perceived water security through NGO activities. Policymaking implications involve employing nature-based slope stabilization and reforestation to boost ground-water recharge, diversifying livelihoods through low-interest microcredit, and institutionalizing gender-sensitive WASH committees to maintain local ownership. All these efforts contribute to SDG 6, as well as enhancing climate resilience among indigenous CHT communities.

In the CHT, households experience large water-collection loads, with an average round-trip fetching time of 7.8 min to obtain drinking water and 19.4 min to obtain domestic water. 60% of respondents are affected by seasonal scarcity, and more than a third of households are in a high-burden category with long distances, long fetching times, and inadequate sanitation facilities. Climate-shock exposure and distance to source proved to be the best predictors of time burden and perceived water adequacy, whereas private toilet ownership reduced the fetching time by half and increased the likelihood of regular water supply more than fourfold. The problem of ethnic differences, which is the most striking in relation to Chakma communities, shows the necessity to plan infrastructure in culturally sensitive ways. These results support the idea of urgent WASH investments: the installation of community tube-wells within 100 m of dwelling clusters, the creation of small water reservoirs, the reinstatement of hilltop forests to enhance groundwater infiltration, and the institutionalization of gender-sensitive governance frameworks. Resilience will be further improved by complementary measures like

low-interest rural credit, disaster-early-warning systems, and participatory watershed management.

This research confirms that tribal communities in the CHT still largely rely on the water streams and canals in the hills as a primary source of water, which is consistent with the previous study on the resource dependency and vulnerability [2, 16]. The acute seasonal scarcity was found between December and April, 72% of households were experiencing extreme or high-water stress in the dry season. Whereas few respondents have already built little irrigation dams (godha), and community tube wells, coverage is still patchy and, in many cases, is not sufficient to meet the demands of drinking and domestic uses, highlighting ongoing infrastructure gaps in this underprivileged area. Although there exist strong national models like BCCSAP and NAPA that place a strong emphasis on community-based adaptation [17], the results of this study reveal an acute gap in implementation. There were financial and institutional disconnects: most communities have not been able to get sufficient funding or coordination to undertake WASH resilience. Respondents emphasized the concept of decentralized governance [18] in accordance with which localized planning and budgetary autonomy are necessary to match the strategic goals with the realities on the ground. This policy practice mismatch destroys climate resilience and continues the water poverty cycle [9].

Water insecurity emerged as a key challenge that has health burdens. Fresh water shortages have a disproportional effect on women, with up to three hours of daily labor on water and survival with poor sanitation, reminiscent of the gendered vulnerabilities elsewhere [11, 19]. To curb this imbalance, gender responsive actions need to be incorporated in WASH policies and community participation mechanisms.

These nature-based and decentralized technological interventions can provide opportunities of resilience. Complementary, rainwater harvesting systems have shown high yield and livelihood gains, doubling the intensity of cropping to 300% in valley lands [20] and continuing 99% dependable by forecasted rainfall conditions [21]. Gravity flow Water Systems Community-size have also minimized the reliance on unsustainable extraction and minimized the workload on women and children [14]. The cost-effective and biogas recovery Floatable WASH platforms also demonstrate the potential of locally adapted designs [12].

Diversification of livelihoods and social capital are identified as important sources of resilience. With regular floods, droughts, and landslides, which have been endorsed by vulnerability mapping [22, 23], agro-jhum systems have been replaced by alternative means of earning incomes, although unequally. Although still a significant number of them adhere to the practice of farming [18, 24], some of them can accept fisheries, small-scale trade, or migrant work [25, 26]. This research modeled regression to understand how better household water security is associated with increased migration rates, indicating that remittances could be used to fund WASH. However, the root cause of poverty and insecurity in land tenure due to unsustainable shifting cultivation makes equitable adaptation hard to achieve [27].

Co-benefits can be magnified by incorporating the principles of the circular economy into the WASH and agricultural waste streams. Compost or bioenergy generation of organic residues can be seen as a critical success factor, based on the examples of the WEEE sector [28, 29] and seed-grant innovations [30, 31], to strengthen livelihoods, promote biodiversity, and reduce GHG emissions. Many communities are interested in

cross-functional community teams to control forest conservation [32, 33] and canal restoration projects, which will reduce waterlogging and improve year-round supply [34, 35].

Collectively, this analysis shows the necessity of integrative, multi-scaled approaches that combine nature-based solutions, decentralized governance, gender-sensitive planning, and circular economy solutions. It will be crucial to strengthen financial flows under the BCCSAP and NAPA, institutional coordination, and empowering local committees to coordinate the tube-well and rainwater systems. The use of longitudinal monitoring of the water quality, health outcomes, and socio-ecological indicators to measure the efficiency of interventions should be used in future research. Bangladesh can forge a resilient course toward the CHT and other at-risk areas by filling policy gaps, harnessing indigenous knowledge, and creating a sustainable way forward.

In addition, these findings emphasize the role of geographic, socio-economic, and cultural aspects in interacting to develop WASH resilience in the CHT. The statistical significance of the negative relationship between the distance to water sources and resilience highlights the multidimensionality of water poverty; physical access to water interacts with livelihood vulnerability to limit adaptive capacity [36–38]. Likewise, the lower resilience of agriculture-dependent households is indicative of the vulnerability of climate-dependent livelihoods, which is consistent with the triple bottom line insights of the necessity to harmonize environmental sustainability, social equity, and economic security [39, 40]. The provision of protective benefits of toilet ownership shows that investments in basic infrastructures can have both short-term health payoffs as well as long-term resilience, supporting the claim that circular economy strategies can create socio-economic and ecological value [31, 41]. The ethnic differences, especially the reduced resilience witnessed among Chakma families, indicate governance and inequality issues that need to be resolved using inclusive policy frameworks. The incorporation of nature-based solutions, including watershed restoration and ecosystem-based sanitation, presents an avenue of addressing climate shocks without violating the practices of the indigenous people and increasing the level of community involvement [42, 43]. Collectively, these results indicate that resiliency in WASH systems cannot only be accomplished through technical intervention but must be systemic and integrate infrastructure, livelihood diversification, and ecosystem stewardship to deliver equitable and sustainable results.

Lastly, this research paper encourages the need to adopt a circular economy at the local level and to provide fundamental health education to everyone. Compost fertilizer prepared through locally made compost latrines can be used to aid the jhum farming since chemical fertilizer is not applied in these areas. It will guarantee not only sanitation with a little water but also enable paddy farming in jhum (because in this process, rainwater is adequate). In addition, groundwater can be utilized during the summer and rainy seasons to achieve healthy drinking water practices. Moreover, health problems can be resolved with basic training in water filtering. Conversely, since the groundwater is present in a shallow form, it can be utilized by renewable sources to supply fundamental cultivation requirements. Finally, there will be No-interest loans to farmers that will motivate them to earn profit. Therefore, these findings can guarantee fundamental WASH to this disadvantaged group with a locally owned solution with good governance and a financial framework.

Through the integration of local priorities with SDG 6 and climate-adaptation targets, the stakeholders are able to provide equitable, sustainable water and sanitation services, minimize labor and health risks, and enhance the adaptive capacity of local people in the CHT.

5 Conclusion

One of the hot topics in Bangladesh is climate change. This paper sheds light on the extreme water and sanitation issues experienced by Chakma, Tripuri, and Bangali families in remote villages of CHT. Hilly aspect and deforestation, as well as irregular rainfall, have extended the period of water collection in round trips up to 30 min and compelled them to resort to insecure surface sources during the summer scramble and during monsoons. Open defecation by a third of the families exacerbates their health issues, and high-interest debt in forms of tobacco advances traps the farmers in the cycle of exploitation, even as agricultural yields are heard to be profitable. The discussion has shown that community-level circular-economy and WASH interventions have several co-benefits:

- (1) Safe fertilizer Compost-latrines would be able to restore vitality in the jhum paddy fields with no chemicals, eliminating the necessity of shifting cultivation, and delaying deforestation.
- (2) Groundwater pumping year-round on solar energy by using renewable water resources can reduce the time spent by women in collecting potable water, as well as eliminate the number of drought-related pollution during floods.
- (3) Simple training on water-filtration will enable cash-strapped households to benefit during rainy seasons by treating shallow groundwater and other surface supplies and reducing incidences of diarrheal diseases.
- (4) Free microcredit to farmers will end the trap of usury lending so that farmers can invest in more resilient infrastructures and diversified livelihoods.

This paper illustrates that the complexity of geographic, socio-economic, and cultural factors influences WASH resilience in CHT. The findings highlight that the reliance on water sources and livelihoods, exposure to climate shocks, and sanitation infrastructure are key determinants of resiliency, with significant differences among ethnic communities. A clear connection of the findings into the framework of the circular economy reveals how resource recovery, waste reduction, and the closed-loop can be used to enhance WASH services and, at the same time, promote environmental sustainability and socioeconomic equity. Integrating nature-based solutions in this model augments adaptive capacity by capitalizing on the services provided by the ecosystem to reduce climate risks.

Governance-wise, there is evidence indicating the necessity of an inclusive and multi-tiered institutional set-up with an emphasis on equity, community involvement, and cross-sectoral coordination. The policymakers need to abandon the compartmentalized approaches of interventions in favor of coherent approaches that would ensure that the principles of the circular economy are consonant with local systems of governance. Enhancing accountability, representation of indigenous people, and strengthening collaboration between the government, civil society, and local actors are critical to come up with a plan of action for operationalizing the resilience strategies. Placing WASH

resilience in the framework of the circular economy and the governance system, this study offers practical recommendations to policymakers and practitioners interested in developing sustainable, equitable, and climate-resilient futures in the Hill Tracts of Bangladesh.

Supplementary Information

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Supplementary Material 1.

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Author contributions

ST generated the idea, designed the methodology, and prepared the draft. MMR conducted the survey and FGD.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study received ethical approval from the University Ethics Committee, Rajshahi University of Engineering & Technology, with administrative endorsement from the Director of Research and Extension, Rajshahi University of Engineering & Technology. The study protocol, participant information sheet, informed consent form, detailed questionnaire, and investigation plan were submitted to the committee for review. Approval was granted under Ref: DRE/8/EC/2026/007. The research was conducted in accordance with the Declaration of Helsinki and applicable national research ethics regulations. Informed consent was obtained from all participants before inclusion in the study. Participants were provided with written information about the study's purpose, procedures, potential risks and benefits, confidentiality protections, and their right to withdraw at any time without penalty. Participants who were able to sign provided written informed consent. For participants who were unable to sign, the ethics committee approved an alternative documented procedure (witnessed verbal consent recorded in writing), and this approved procedure was followed and recorded. Copies of signed consent forms and records of any approved alternative consent procedures are retained by the investigators and were submitted to the University Ethics Committee as part of the approval process.

Consent for publication

Informed consent was obtained from all participants to publish the information/images in an online open-access publication, with all data anonymized prior to publication.

Competing interests

The authors declare no competing interests.

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